# Stochastic Finance and Risk Modelling

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# 3 - One-period model of financial market

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# Outline

Market, assets and portfolio

Arbitrage and FTAP

Payoffs, replicable payoffs, derivatives

Complete market

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# Assets I

- Consider a market of one period (or equivalently two dates) : 0 now, and $T = 1$ future.
- Suppose that $d + 1$ assets are traded in the market, and denote by $S_t^i$ the price of asset $i$ at time $t \in \{0, 1\}$.

**Definition (Riskless asset)**

Suppose asset 0 is riskless :

$$S_0^0 > 0, \quad S_1^0 = S_0^0(1 + r),$$

where $r$ is a constant and $r > -1$. We will often choose $S_0^0 = 1$.

- Interpretation : loan/saving with interest rate $r$

# Assets II

- Suppose that the other $d$ assets are risky. Their initial price is a constant $S_0 = (S_0^1, \ldots, S_0^d) \in \mathbb{R}_+^d$ and their price at time 1 (at the end of the period) is a random variable denoted by $S_1 = (S_1^1, \ldots, S_1^d)$.
- We work on the probability space $(\Omega, \mathcal{F}_1, \mathbb{P})$. Each $\omega \in \Omega$ is a state of nature.

**Definition (Risky assets)**

The price $S_1^i$ is a r.v. on $(\Omega, \mathcal{F}_1, \mathbb{P})$, i.e. $S_1^i$ is $\mathcal{F}_1$-measurable and takes values in $\mathbb{R}_+$ for all $i \in \{1, \ldots, d\}$.

- $S_1 = (S_1^1, \ldots, S_1^d)$ is a random vector taking values in $\mathbb{R}_+^d$.

# Assets III

How to construct $(\Omega, \mathcal{F}_1)$ ?

- $\Omega := \mathbb{R}_+^d$ ;
- $\mathcal{F}_1 := \mathcal{B}(\mathbb{R}_+^d)$ ;
- $S_1 : \Omega \to \mathbb{R}_+^d$ is defined by $S_1(\omega) := \omega$ for all $\omega \in \mathbb{R}_+^d$ ;
- Here $\omega$ corresponds to a possible price trajectory.

# Portfolio

- An investor chooses an investment strategy by buying or selling assets.

## Definition

A portfolio is identified by $\xi^0 \in \mathbb{R}$ and $\xi = (\xi^1, \dots, \xi^d) \in \mathbb{R}^d$ representing the number of shares of the assets hold by the investor in the riskless asset and in the risky assets respectively.

- $(\xi^0, \xi) \in \mathbb{R}^{d+1}$
- $\xi^i > 0$ : buy (or *long sale*) ; $\xi^i < 0$ : sell (or *short sale*).
- Initial value of the portfolio :

$$V_0^{\xi^0,\xi} := \xi^0 S_0^0 + \xi \cdot S_0 = \xi^0 S_0^0 + \sum_{i=1}^{d} \xi^i S_0^i.$$

- Final value of the portfolio :

$$V_1^{\xi^0,\xi} = \xi^0 S_1^0 + \xi \cdot S_1 = \xi^0 S_0^0 (1+r) + \sum_{i=1}^{d} \xi^i S_1^i.$$

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# Arbitrage

## Definition

We say that a portfolio $(\xi^0, \xi)$ yields an arbitrage opportunity if

(i) $V_0^{\xi^0,\xi} = 0$,

(ii) $V_1^{\xi^0,\xi} \geq 0$ $\mathbb{P}$-a.s.,

(iii) $\mathbb{P}[V_1^{\xi^0,\xi} > 0] > 0$.

- The existence of an arbitrage is perceived as a market inefficiency, which will be exploited and quickly disappears.
- When there is no arbitrage, the market is called arbitrage free or efficient.

# Arbitrage and strategy

From $V_0^{\xi^0,\xi} = \xi^0 S_0^0 + \xi \cdot S_0 = 0$, we deduce $\xi^0 = -\xi \cdot \dfrac{S_0}{S_0^0}$. Then

$$V_1^{\xi^0,\xi} = \xi^0 S_1^0 + \xi \cdot S_1 = -\xi \cdot \frac{S_0}{S_0^0} S_1^0 + \xi \cdot S_1 = \xi \cdot S_1 - (1+r)\xi \cdot S_0.$$

## Proposition

*The market admits an arbitrage opportunity iff there exists a vector $\xi \in \mathbb{R}^d$ s.t.*

(i) $\xi \cdot S_1 \geq (1+r)\xi \cdot S_0$, $\mathbb{P}$-*a.s.*,

(ii) $\mathbb{P}[\xi \cdot S_1 > (1+r)\xi \cdot S_0] > 0$.

- When $d = 1$, (i) is equivalent to $\dfrac{S_1^1 - S_0^1}{S_0^1} \geq r$ if $\xi > 0$ or $\dfrac{S_1^1 - S_0^1}{S_0^1} \leq r$ if $\xi < 0$.
- Interpretation for $\xi > 0$ : an investment in risky assets cannot "systematically " beat the riskless asset without creating an arbitrage.

# Discounted price

## Definition

Denote by $\tilde{S}^i$ the discounted price of asset $i$, i.e.

$$\tilde{S}_t^i = \frac{S_t^i}{S_t^0} \text{ for } t \in \{0, 1\}.$$

We also denote by $\tilde{V}^{\xi^0,\xi}$ the discounted value of the portfolio $(\xi^0, \xi)$

$$\tilde{V}_t^{\xi^0,\xi} = \frac{V_t^{\xi^0,\xi}}{S_t^0} \text{ for } t \in \{0, 1\}.$$

# Discounted price

Let $\Delta\tilde{S}_1^i = \tilde{S}_1^i - \tilde{S}_0^i$ for $i \in \{0, \dots, d\}$ and $\Delta\tilde{S}_1 = (\Delta\tilde{S}_1^1, \dots, \Delta\tilde{S}_1^d)$.
Note that $\Delta\tilde{S}_1^0 = 0$.

## Proposition

*The market admits an arbitrage opportunity iff there exists a vector $\xi \in \mathbb{R}^d$ s.t.*

(i) $\xi \cdot \Delta\tilde{S}_1 \geq 0$, $\mathbb{P}$-*a.s.*,

(ii) $\mathbb{P}[\xi \cdot \Delta\tilde{S}_1 > 0] > 0$.

# Martingale measure

## Definition

A probability $\mathbb{Q}$ on $(\Omega, \mathcal{F}_1)$ is called risk-neutral or a martingale measure if

$$S_0^i = \mathbb{E}^{\mathbb{Q}}\left[\frac{S_1^i}{1+r}\right], \quad i = 1, \ldots, d.$$

Recall that

$$\tilde{S}_0^i = \frac{S_0^i}{S_0^0} \quad \tilde{S}_1^i = \frac{S_1^i}{S_0^0(1+r)}.$$

- The notation "martingale" comes from $\tilde{S}_0^i = \mathbb{E}^{\mathbb{Q}}\left[\tilde{S}_1^i\right]$.
- The notation "risk-neutral" comes from $\mathbb{E}^{\mathbb{Q}}\left[\frac{S_1^i - S_0^i}{S_0^i}\right] = r$.

# Fondamental Theorem of Asset Pricing (FTAP)

**Theorem**

*The market is arbitrage-free iff there exists a martingale measure $\mathbb{Q}$ that is equivalent to $\mathbb{P}$.*

# Return

Let $(\xi^0, \xi)$ identify a portfolio of value $V_t^{\xi^0,\xi} = \xi^0 S_t^0 + \xi \cdot S_t$, $t = 0, 1$.

## Definition

If $V_0^{\xi^0,\xi} \neq 0$, the portfolio's return is defined by

$$\rho(V^{\xi^0,\xi}) := \frac{V_1^{\xi^0,\xi} - V_0^{\xi^0,\xi}}{V_0^{\xi^0,\xi}} = \frac{\xi^0(S_1^0 - S_0^0) + \xi \cdot (S_1 - S_0)}{\xi^0 S_0^0 + \xi \cdot S_0}.$$

- Return of riskless asset : $\rho(S^0) = \rho(V^{1,0}) = r$
- Return of risky asset : $\rho(S^i) = \rho(V^{0,1_i}) = \dfrac{S_1^i - S_0^i}{S_0^i} = \dfrac{S_1^i}{S_0^i} - 1$, where $1_i$ is the $\mathbb{R}^d$ vector with all entries equal to zero expect entry $i$ equals to 1.

# Return

- Let $w_k := \dfrac{\xi^k S_0^k}{\sum_{i=0}^d \xi^i S_0^i} = \dfrac{\xi^k S_0^k}{V_0^{\xi^0,\xi}}$ be the fraction of initial wealth invested in asset $k$.
- Return of portfolio in assets : $\rho(V^{\xi^0,\xi}) = \displaystyle\sum_{k=0}^{d} w_k \rho(S^k)$.

# Space of returns

Consider a portfolio determined by $(\xi^0, \xi)$ whose value process is given by $V_t^{\xi^0,\xi} = \xi^0 S_t^0 + \xi \cdot S_t,\ t = 0, 1.$

**Proposition**

*If the market is arbitrage-free and $\mathbb{Q}$ is a martingale measure equivalent to $\mathbb{P}$, then $\mathbb{E}^{\mathbb{Q}}\left[\rho(V^{\xi^0,\xi})\right] = r$.*

- Under equivalent martingale measures, all the portfolios have the same return.

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# Payoff

## Definition

A payoff is identified by a r.v. on $(\Omega, \mathcal{F}_1, \mathbb{P})$.

- Interpretation : payment depending on the scenario $\omega$ at time $t = 1$.

## Definition

The space $\{\xi^0 S_1^0 + \xi \cdot S_1 : \xi^0 \in \mathbb{R}, \xi \in \mathbb{R}^d\}$ is called the space of replicable payoffs.

- Interpretation : payoff that can be reproduced by a portfolio.

## Proposition

*Assume that the market is arbitrage-free and a payoff $H$ is replicable. For any two replicating portfolios $(\xi^0, \xi)$ and $(\zeta^0, \zeta)$, they have the same value at time $t = 0$.*

# Derivatives I

## Definition

Assume that $\mathcal{F}_1 = \sigma(S_1^0, \ldots, S_1^d)$. A derivative of payoff $H$ is a function of the prices $S_1^0, \ldots, S_1^d$, i.e.

$$H = f(S_1^0, \ldots, S_1^d)$$

for some measurable function $f : \mathbb{R}^{d+1} \to \mathbb{R}$.

- Forward contract : We call a forward contract on the asset $S^i$ the agreement at $t = 0$ between the buyer and the seller s.t. the buyer will buy at $t = 1$ one unit of the asset $S^i$ with the fixed price $K$ (at time 0). The corresponding payoff is therefore written

$$H^F = S_1^i - K.$$

# Derivatives II

- Call option : We call a call option on the asset $S^i$ the agreement at $t = 0$ between the buyer and the seller s.t. the buyer has the right, but not the obligation, to buy at $t = 1$ one unit of the asset $S^i$ with the fixed price $K$ (called strike and decided at time 0). The corresponding payoff is therefore written

$$H^C = \max\left(S_1^i - K, 0\right) = \left(S_1^i - K\right)^+.$$

- Put option : Similarly,

$$H^P = \max\left(K - S_1^i, 0\right) = \left(K - S_1^i\right)^+.$$

- Basket option, straddle, butterfly spread, etc.
- **In what follows**, we focus on the positive payoffs.

# Arbitrage-free price

Denote by $\mathcal{M}$ the market consisting of the assets $S^0, \ldots, S^d$. Consider a derivative of payoff $H$.

## Definition

We call $p$ the arbitrage-free price of $H$ if the extended market $\mathcal{N}$ is still arbitrage-free, where $\mathcal{N} := \mathcal{M} \cup \{S^{d+1}\}$ with

$$S_0^{d+1} = p, \quad S_1^{d+1} = H.$$

# Arbitrage-free price and martingale measure

Consider a derivative of payoff $H$. Assume that $S_0^0 = 1$.

**Theorem**

*Let the market $\mathcal{M}$ be arbitrage-free. Then the set of arbitrage-free prices of $H$ is given by*

$$\left\{ \mathbb{E}^{\mathbb{Q}} \left[ \frac{H}{S_1^0} \right] : \mathbb{Q} \text{ martingale measure } \mathbb{Q} \sim \mathbb{P} \text{ s.t. } \mathbb{E}^{\mathbb{Q}} \left[ \left| \frac{H}{S_1^0} \right| \right] < \infty \right\}.$$

# Replicable derivative

## Definition

A derivative $H$ is called replicable if its payoff can be reproduced by a portfolio $(\xi^0, \xi) \in \mathbb{R}^{d+1}$, i.e. $\exists(\xi^0, \xi)$ s.t. $H = V_1^{\xi_0,\xi} = \xi^0 S_1^0 + \sum_{i=1}^{d} \xi^i S_1^i$.

The portfolio $(\xi^0, \xi)$ is called a replicating portfolio of $H$.

Then $\mathbb{E}^{\mathbb{Q}}\left[\frac{H}{S_1^0}\right] = \mathbb{E}[\tilde{V}_1^{\xi_0,\xi}] = \tilde{V}_0^{\xi_0,\xi}$ for every replicating portfolio $(\xi^0, \xi) \in \mathbb{R}^{d+1}$.

## Theorem

*A derivative is replicable if and only if there is a unique arbitrage-free price.*

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**Definition**

A market is said to be complete if all the payoffs are replicable.

**Theorem**

*An arbitrage-free market is complete iff there exists a unique equivalent martingale measure.*